

yc4384_P8122_HW3

yc4384_Yangyang_Chen

2024-10-11

Data Preparation for Mice Study

```
set.seed(124)
n_mice = 16
p_obesity = 1/5
obesity = rbinom(n_mice, 1, p_obesity)
theta_0 = 1/2
theta_1 = -1/5
p_light = theta_0 + theta_1 * obesity
light_exp = rbinom(n_mice, 1, p_light)
beta_0 = 110
beta_1 = 20
beta_2 = -5
sigma_Y = 1
mu_glucose = beta_0 + beta_1 * obesity + beta_2 * light_exp
glucose = rnorm(n_mice, mu_glucose, sigma_Y)

mice_data = data.frame(
  mouse_id = 1:n_mice,
  obesity = obesity,
  light_exposure = light_exp,
  glucose = glucose
)

mice_data |> head()
```

```
##   mouse_id obesity light_exposure  glucose
## 1         1       0               1 104.8490
## 2         2       0               1 103.7769
## 3         3       0               1 104.1312
## 4         4       0               0 108.9575
## 5         5       0               1 103.8964
## 6         6       0               1 105.4442
```

Problem 1: Interpretation of parameters:

p : Probability that the baseline covariate C (obesity) equals 1, indicating the mouse is obese (Bernoulli distribution).

θ_0 : Probability of assigning mice to light exposure when not obese ($C = 0$).

θ_1 : Probability of assigning obese mice to light exposure is reduced by $1/5$ compared to non-obese mice.

β_0 : Average baseline glucose level when the mouse is not obese and unexposed to light.

β_1 : A unit increase in obesity results in an average increase of 20 units in glucose levels, holding other factors constant.

β_2 : Exposure to light reduces glucose levels by 5 units on average, holding other factors constant.

Problem 2: Marginal and Conditional PACE

The Marginal PACE is:

$$E[Y_1] - E[Y_0] = \sum E[Y|A = 1, C = c]P(C = c) - \sum E[Y|A = 0, C = c]P(C = c)$$

The Conditional PACE is:

$$E[Y_1|C = c] - E[Y_0|C = c] = \beta_2 = -5$$

I. Under the following assumptions Marginal PACE can be identified:

1) Consistency:

$$Y = Y_1 \text{ if treated, and } Y = Y_0 \text{ if not treated.}$$

2) Exchangeability:

$$Y_1, Y_0 \perp A|X.$$

3) Positivity:

$$0 < P(A = 1|X) < 1.$$

4) SUTVA: No interference between units, and the treatment is well-defined for all units.

II. Under the following assumptions Conditional PACE can be identified:

1) Conditional Consistency:

$$Y = Y_1 \text{ if } A = 1, \quad Y = Y_0 \text{ if } A = 0, \quad \text{given } X.$$

2) Conditional Exchangeability:

$$Y_1, Y_0 \perp A|X = x.$$

3) Conditional Positivity:

$$0 < P(A = 1|X = x) < 1.$$

4) SUTVA: No interference between units, and the treatment is well-defined for all units.

Problem 3: G-formula for Randomized and Observational Studies

g-formula for the randomized study: $\sum_c E[Y|A = 1, C = c]Pr(C = c) - \sum_c E[Y|A = 0, C = c]Pr(C = c) = E[Y|A = 1] - E[Y|A = 0]$

g-formula for the observational study: $\sum_c E[Y|A = 1, C = c]Pr(C = c) - \sum_c E[Y|A = 0, C = c]Pr(C = c)$ under NUCA.

Randomization ensures unconfoundedness, meaning that the distribution of C is independent of A (the treatment assignment). As a result, the g-formula simplifies to $E[Y|A = 1] - E[Y|A = 0]$, as no adjustment for confounders is necessary. However, in an observational study, A is influenced by C, meaning that the relationship between the treatment and outcome is confounded. In such cases, the g-formula cannot be simplified without appropriately accounting for the confounders.

Problem4: Estimate and confidence interval of $E[Y|A = 1] - E[Y|A = 0]$

The 95% confidence interval can be calculated by $E[Y|A = 1] - E[Y|A = 0] \pm z\sqrt{\frac{S_1^2}{N_1} + \frac{S_0^2}{N_0}}$, where z is the critical value from the standard normal distribution, S_1^2 and S_0^2 are the variances of the potential outcomes in $A = 1$ and $A = 0$ groups, and N_1 and N_0 are the sizes of $A = 1$ and $A = 0$ groups:

```
# Split Y into treatment and control groups
Y_1 = glucose[light_exp == 1]
Y_0 = glucose[light_exp == 0]

# Use pipeline to compute mean glucose levels and causal effect estimate
mean_Y_1 = Y_1 |> mean()
mean_Y_0 = Y_0 |> mean()

# Causal effect estimate
diff_estimate = mean_Y_1 - mean_Y_0

# Critical value for 95% CI
z_crit = qnorm(1 - 0.05 / 2)

# Variances and sample sizes
var_Y_1 = Y_1 |> var()
var_Y_0 = Y_0 |> var()
n_1 = Y_1 |> length()
n_0 = Y_0 |> length()

# Standard error
se_diff = (var_Y_1/n_1 + var_Y_0/n_0) |> sqrt()

# Confidence interval
ci_lower = diff_estimate - z_crit * se_diff
ci_upper = diff_estimate + z_crit * se_diff
```

Therefore, the estimate of causal effect is $E[Y|A = 1] - E[Y|A = 0] = -12.57$. The 95% confidence interval is $(-19.68, -5.45)$, which excludes zero, indicating a causal effect between the bright and dark light groups in obesity. Therefore, the result is statistically significant at the 95% confidence level.

Additionally, the point estimate suggests that, on average, the outcome Y is 12.57 mg/dL lower in bright light group compared to dark light group.

5. Estimate and confidence interval of $E[Y_1] - E[Y_0]$

The difference $E[Y_1] - E[Y_0]$ is calculated using the g-formula from Problem 3: $E[Y_1] - E[Y_0] = \sum_c E[Y|A = 1, C = c]Pr(C = c) - \sum_c E[Y|A = 0, C = c]Pr(C = c)$

```
p_C0 = obesity |> mean()
p_C1 = 1 - p_C0

# Mean outcomes for different combinations of A and C
mean_Y_0_0 = glucose[light_exp == 0 & obesity == 0] |> mean()
mean_Y_0_1 = glucose[light_exp == 0 & obesity == 1] |> mean()
mean_Y_1_0 = glucose[light_exp == 1 & obesity == 0] |> mean()
mean_Y_1_1 = glucose[light_exp == 1 & obesity == 1] |> mean()
```

Upon noticing that no obese mice were assigned to the bright light group, this violates the probabilistic assumption. Therefore, we will exclude the obese mice and consider only the non-obese group.

```
# g-formula estimates
E_Y1 = p_CO * mean_Y_1_0
E_Y0 = p_CO * mean_Y_0_0

# Difference E[Y1] - E[Y0]
diff = E_Y1 - E_Y0
```

Therefore, $E[Y_1] - E[Y_0] = -0.97$

```
# Critical value for 95% CI
z = qnorm(1 - 0.05 / 2)

# Sample sizes for A=1 and A=0, C=0
n_1_0 = sum(light_exp == 1 & obesity == 0)
n_0_0 = sum(light_exp == 0 & obesity == 0)

# Variances for A=1 and A=0, C=0
var_Y_1_0 = glucose[light_exp == 1 & obesity == 0] |> var()
var_Y_0_0 = glucose[light_exp == 0 & obesity == 0] |> var()

# Standard error
se_diff = (var_Y_1_0 / n_1_0 + var_Y_0_0 / n_0_0) |> sqrt()

# Confidence interval
ci_lower = diff - z * se_diff
ci_upper = diff + z * se_diff
```

The 95% confidence interval is (-2.14, 0.2).

The point estimate indicates that, on average, the outcome Y is 0.97 mg/dL higher in the bright light group compared to the dark light group among non-obese mice in this observational study. However, due to the violation of probabilistic assumptions, we cannot infer the causal effect for obese mice. Since the 95% confidence interval includes zero, the point estimate is not statistically significant at the 95% confidence level.

The difference between the point estimates in Problems (4) and (5) arises because Problem (5) accounts for the adjustment of confounding bias.

6. Assumptions of estimate $E[Y_1] - E[Y_0]$ using linear regression with continuous covariates

In addition to consistency, SUTVA, exchangeability, positivity, and NUCA assumption, we need to assume the following:

- No measurement error: All variables need to be correctly measured.
- No model misspecification: All models need to be correctly specified. In the linear regression, the relationship between Y , A , and covariates should be adequately approximated by a linear model.